1. **HTML elements which don't need to be closed, such elements are known as ______________________**

a) Empty Elements b) Void Elements

c) Blank Elements d) Open Elements

2. **The total no. of Host in the network in which the given IP- 200.10.20.40/25 resides will be**

a) 32 b) 34

c) 64 d) 128

3. **FTP is considered an __________________ protocol.**

a) Presentation Layer b) Application Layer

c) Network Layer d) Physical Layer

4. **An identical protocol to GBN, in which buffers used at the receiver and the sender end maintains a window of same size is called ___________________**

a) Selective Repeat Protocol b) Stop and Wait protocol

c) Cumulative Repeat protocol d) None of these

5. **___________ are used instead of an A record when there is an alias.**

a) NS record b) TXT record
c) CNAME record d) Other

6. **Which attribute is used to provide a unique name to an HTML element?**

a) ID b) CLASS
c) TYPE d) None of these

7. **The total no. of Host in the network in which the given IP- 200.10.20.40/24 resides will be**

a) 124 b) 256
c) 258 d) 128

8. **What is the correct syntax to write an HTML comment?**

(a) <!-- Comment --> (b) // Comment

(c) # Comment (d) /* Comment */

9. **A domain name is made of multiple parts, called _________________**

i. labels ii. Mapping
iii. Options iv. Other

<u>5 marks</u>

1. What is NAT? How does it work?

NAT (Network Address Translation) is a method used in routers to **map multiple private IP addresses to a single public IP address** before data is sent over the Internet.

Types of NAT:

Type	Description
Static NAT	One-to-one mapping between private and public IPs.
Dynamic NAT	Private IPs are mapped to a pool of public IPs.
PAT (Port Address Translation) / NAT Overload	Many private IPs share one public IP using port numbers.

Outbound Communication (from LAN to Internet):

- A device in the private network sends a request to an external server.
- The router replaces the **source IP** with its **public IP** and sends the packet to the internet.
- When the response comes back, the router uses a **translation table** to forward it to the correct internal device.

2. Write Short notes on

i. Distance Vector Routing

ii. RARP

Distance Vector Routing

- **Definition:** A routing protocol where each router shares its routing table with its directly connected neighbors periodically.
- **How it Works:** Routers use the Bellman-Ford algorithm to calculate the shortest path by counting the number of hops.
- **Updates:** Routers update their routing tables based on the information received from neighbors.
- **Advantages:** Simple and easy to implement.
- **Disadvantages:** Slow convergence, can create routing loops.
- **Examples:** RIP (Routing Information Protocol).

RARP (Reverse Address Resolution Protocol)

- **Definition:** A protocol used to map a **MAC address to an IP address**.
- **Use Case:** Used by diskless computers at boot time to obtain their IP address from a RARP server.
- **Function:** Opposite of ARP (Address Resolution Protocol), which maps IP to MAC.
- **Limitation:** Obsolete today; replaced by **BOOTP** and **DHCP** which provide more features.

3. What are the different error control technique is used in Data link layer? What are the limitations we can find in S**top-and- wait** protocol?

Error Control Techniques in the Data Link Layer:

a. Error Detection Techniques:

- **Parity Check:** Adds a parity bit (even/odd) to detect single-bit errors.
- **Checksum:** Sums the data segments and appends the result to detect errors.
- **Cyclic Redundancy Check (CRC):** Uses polynomial division to detect burst errors with high accuracy.

b. Error Correction Techniques:

- **Automatic Repeat Request (ARQ):** If an error is detected, the receiver requests retransmission.

  Types of ARQ protocols:

  - **Stop-and-Wait ARQ**
  - **Go-Back-N ARQ**
  - **Selective Repeat ARQ**
- **Forward Error Correction (FEC):** Adds redundant bits so the receiver can correct errors without retransmission.

Limitations of Stop-and-Wait Protocol:

Inefficiency for High Latency Links: Sender must wait for acknowledgment after every frame, causing idle time on the link.

Low Throughput: Only one frame is in transit at any time, underutilizing available bandwidth.

Wasted Bandwidth: Even if the link is capable of handling multiple frames, it remains idle until an acknowledgment is received.

Not Suitable for High-Speed Networks: The protocol cannot handle large volumes of data efficiently due to its simple one-at-a-time mechanism.

Delay in Transmission: Causes high delay especially when the round-trip time is long.

4. What is Search Engine Optimization (SEO)?

SEO or Search Engine Optimization is defined as the **process of improving (optimising) the visibility of a website/webpage on Search Engines**, such as Google, Bing, etc.

5. Why do we need SEO?

We need Search Engine Optimization for the following aspects:

- To improve the quality of our website

- To increase web traffic
- To increase visibility
- To enhance user experience
- To gain a competitive advantage
- For analysis and monitoring of the website

6. How does SEO work?

- **Crawling:** Search engine bots (crawlers) explore the web, discovering new pages and updating their indexes.
- **Indexing:** The indexed content is stored in a search engine's database, making it searchable.
- **Ranking:** Search engines use complex algorithms to determine the order of search results, prioritizing relevant and authoritative pages.
- **Keyword Research:** Identifying the terms and phrases your target audience uses to search for information related to your business.
- **On-page Optimization:** Optimizing elements within your website's pages, such as title tags, meta descriptions, and content.
- **Off-page Optimization:** Building quality backlinks from other websites, enhancing your website's authority and credibility.
- **Technical SEO:** Ensuring your website is technically sound, including factors like page speed, mobile-friendliness, and proper URL structure.

7. What is Web Crawler?
A web crawler, crawler or web spider, is a computer program that's used to search and automatically index website content and other information over the internet.
Ex. Amazonbot is the Amazon web crawler, Googlebot is the crawler for Google's search engine, Bingbot is Microsoft's search engine crawler for Bing etc.

8. How does Web Crawler works?

- **Starting Point:** The crawler begins at a designated starting URL.
- **Following Links:** It then follows hyperlinks on that page, visiting and analyzing the linked pages.
- **Extracting Information:** During its visits, it extracts data like text, images, keywords, and meta tags from each page.
- **Indexing:** The extracted information is then stored in a database, creating an index that search engines use to find relevant information when users search.
- **Regular Updates:** Crawlers are often programmed to revisit websites periodically to update their index as websites change.

9. Why web crawlers are important for SEO?

- **Discovery and Indexing:** Web crawlers systematically explore the internet, following links and discovering new and updated pages.
- **Rankings:** The information gathered by crawlers, including keywords, relevance, and user engagement metrics, influences how search engines determine a website's ranking in search results.
- **Content Visibility:** If a website is not crawled, it will not be indexed, and it will not appear in search results, making it practically invisible to potential visitors.
- **Crawl Budget:** Search engines like Google have a limited crawl budget, meaning they can only crawl a certain number of pages on a website within a given time period.
- **Technical SEO:** Web crawlers are essential for identifying technical issues on a website, such as broken links, duplicate content, and redirects, which can negatively impact SEO.

10. What is the differences between Web Crawling and Web Scraping?

Feature	**Web Crawling**	**Web Scraping**
Focus	Discovering URLs and links	Extracting specific data from URLs
Purpose	Indexing the web for search engines or archiving	Collecting data for analysis, research, or other purposes
Output	Primarily lists of URLs and links	Data extracted from the target
Process	Starts with a seed URL and follows links	Involves parsing HTML to extract specific elements
Tools	Crawlers, spiders, or bots	Scraping tools, libraries, and programming languages

1. Describe Client Server Architecture of the Internet.

The **Client-Server Architecture** of the Internet is a fundamental model for designing networked applications and systems, where tasks and services are divided between **clients** (requesters of services) and **servers** (providers of services).

<u>Client</u><u>:</u> A client is a device or application that initiates communication by sending a request to the server.

<u>Server</u><u>:</u> A server is a powerful system that listens for client requests and responds accordingly. It provides resources or services such as web hosting, file storage, or database management.

<u>Working of Client-Server Architecture on the Internet:</u>

Client sends a request:

- For example, when a user types a URL in a browser, the browser (client) sends an HTTP request to the web server.

Server processes the request:

- The server receives the request, fetches or generates the necessary data, and prepares a response.

Server sends response:

- The server sends the data back to the client over the Internet using appropriate protocols like HTTP/HTTPS.

Client displays the result:

- The client renders the data to the user.

2. What is Intranet? Explain its uses.

An **Intranet** is a **private network** that is accessible only to an organization's staff. It uses internet protocols (like TCP/IP, HTTP) but is **restricted to internal use** within an organization. Unlike the public Internet, an intranet is **secured, controlled, and not accessible to outsiders** without proper authorization.

<u>Uses of Intranet:</u>

Internal Communication:

- Facilitates seamless communication through internal messaging, announcements, and news updates.

Document Management:

- Employees can store, access, and share documents, manuals, and reports in a centralized location.

Collaboration:

- Tools like calendars, task managers, and discussion forums help teams work together efficiently.

Training and Education:

- Hosts training materials, e-learning modules, and company policies for employee development.

Process Automation:

- Internal workflows like leave applications, expense claims, and approvals are automated.

Employee Self-Service:

- Employees can update personal information, check salaries, apply for leaves, etc.

Improved Productivity:

- Reduces the time spent on searching for information and facilitates faster decision-making.

3. What is DNS? Explain it with an example.

DNS stands for **Domain Name System**. DNS translates **human-readable domain names** into **IP addresses** that computers use to identify each other on the network.

How DNS Works:

User Enters URL:

- You type www.example.com into your web browser.

DNS Query Sent:

- Your computer sends a request to a **DNS resolver.**

Resolver Checks Cache:

- If the resolver has the IP address for the domain in its cache, it returns it immediately.
- If not, it queries other DNS servers:
    - **Root DNS Server** → directs to Top-Level Domain (TLD) servers like .com
    - **TLD Server** → directs to the domain's **Authoritative Name Server**
    - **Authoritative DNS Server** → provides the actual IP address for www.example.com

Browser Connects to Website:

- Your browser uses the IP address to connect to the web server and load the website.

Example:

You type: www.google.com

DNS resolves it to: 142.250.64.100

Your browser then connects to this IP to display the Google homepage.

4. Differentiate between ARP and RAR. How does DNS increase web performance?

Difference Between ARP and RARP:

Aspect	ARP (Address Resolution Protocol)	RARP (Reverse Address Resolution Protocol)
Purpose	Resolves **IP address** to **MAC address**	Resolves **MAC address** to **IP address**
Direction	IP → MAC	MAC → IP
Used By	Mostly by hosts to find the physical address on LAN	Used by diskless computers to get an IP from the server
Operation	Broadcasts a request asking "Who has this IP?"	Broadcasts a request asking "Who am I?"
Availability	Widely used in modern networks	Rarely used, now mostly replaced by **BOOTP** or **DHCP**
Layer	Works at the **Data Link Layer (Layer 2)**	Also works at the **Data Link Layer (Layer 2)**

How Does DNS Increase Web Performance?

DNS improves web performance in several ways:

DNS Caching:

- Resolved IP addresses are stored temporarily at multiple levels. This reduces the need for repeated lookups, speeding up website access.

Load Balancing:

- DNS can return different IP addresses for the same domain, distributing user requests across multiple servers. This improves speed and prevents server overload.

Geographical Redirection (Geo-DNS):

- DNS can direct users to the nearest server (CDN) based on their location. Reduces latency and improves page load times.

Failover and Redundancy:

- DNS can be configured to redirect traffic to backup servers if the primary one fails. Ensures high availability and better user experience.

Content Delivery Networks (CDNs):

- CDNs rely on smart DNS routing to serve content from edge locations, speeding up access to images, videos, etc.

5. Create a HTML file to demonstrate Ordered Lists and unordered List.

Here's a complete HTML file that demonstrates both **ordered lists** (<ol>) and **unordered lists** (<ul>):

```
<!DOCTYPE html>
<html>
<head>
  <title>Ordered and Unordered Lists Example</title>
</head>
<body>
  <h1>HTML Lists Demonstration</h1>

  <h2>Unordered List</h2>
  <p>This is an example of an unordered list:</p>
  <ul>
    <li>Apples</li>
    <li>Bananas</li>
    <li>Oranges</li>
    <li>Mangoes</li>
  </ul>

  <h2>Ordered List</h2>
  <p>This is an example of an ordered list:</p>
  <ol>
    <li>Wake up</li>
    <li>Brush teeth</li>
    <li>Have breakfast</li>
    <li>Start work</li>
  </ol>
```

```
</body>
</html>
```

6. Create a HTML file to design a table to show current month calendar.

Here's a complete HTML file that creates a **calendar table for the current month (May 2025)**:

```
<!DOCTYPE html>
<html>
<head>
  <title>May 2025 Calendar</title>
</head>
<body>
  <h2>May 2025 Calendar</h2>

  <table>
    <tr>
      <th>Sun</th>
      <th>Mon</th>
      <th>Tue</th>
      <th>Wed</th>
      <th>Thu</th>
      <th>Fri</th>
      <th>Sat</th>
    </tr>

    <tr>
      <td></td>
      <td></td>
      <td></td>
      <td></td>
      <td>1</td>
      <td>2</td>
      <td>3</td>
    </tr>

    <tr>
      <td>4</td>
      <td>5</td>
      <td>6</td>
      <td>7</td>
      <td>8</td>
      <td>9</td>
      <td>10</td>
    </tr>
```

```
    <tr>
      <td>11</td>
      <td>12</td>
      <td>13</td>
      <td>14</td>
      <td>15</td>
      <td>16</td>
      <td>17</td>
    </tr>

    <tr>
      <td>18</td>
      <td>19</td>
      <td>20</td>
      <td>21</td>
      <td>22</td>
      <td>23</td>
      <td>24</td>
    </tr>

    <tr>
      <td>25</td>
      <td>26</td>
      <td>27</td>
      <td>28</td>
      <td>29</td>
      <td>30</td>
      <td>31</td>
    </tr>
  </table>
</body>
</html>
```

7. Create a registration form using HTML with Text input, textarea,radio button,checkbox and submit button.

Here's a complete **HTML registration form:**

```
<!DOCTYPE html>
<html>
<head>
  <title>Registration Form</title>

</head>
<body>
  <h2>Registration Form</h2>
  <form action="#" method="post">
```

```
    <label for="fullname">Full Name:</label>
    <input type="text" id="fullname" name="fullname" required>

    <label for="address">Address:</label>
    <textarea id="address" name="address" rows="4" required></textarea>

    <label>Gender:</label>
    <div class="radio-group">
      <label><input type="radio" name="gender" value="Male"> Male</label>
      <label><input type="radio" name="gender" value="Female"> Female</label>
      <label><input type="radio" name="gender" value="Other"> Other</label>
    </div>

    <label>Hobbies:</label>
    <div class="checkbox-group">
      <label><input type="checkbox" name="hobbies" value="Reading"> Reading</label>
      <label><input type="checkbox" name="hobbies" value="Traveling"> Traveling</label>
      <label><input type="checkbox" name="hobbies" value="Music"> Music</label>
    </div>
    <input type="submit" value="Register">
  </form>
</body>
</html>
```

8. What is Extranet? Explain its uses.

An **Extranet** is a **private network** that allows **authorized external users** (such as customers, vendors, partners, or suppliers) to access specific resources of an organization's **intranet** over the **Internet**, usually through secure means like VPNs or login portals.

<u>Uses of Extranet:</u>

Business-to-Business Collaboration:
- Companies can coordinate with suppliers or distributors to manage inventory, order status, and deliveries.

Customer Support Portals:
- Businesses provide product documentation, support tickets, or FAQs to clients.

Project Management:
- Multiple organizations working on a joint project can share documents, deadlines, and updates.

Secure File Sharing:
- Share sensitive documents (contracts, invoices, reports) securely with partners or clients.

Vendor Access:
- Vendors can track their order history, delivery status, or billing through a controlled interface.

9. What is Subdomain? Explain it with an example.

A **subdomain** is a **prefix** added to a domain name to organize or separate parts of a website. It is used to create a **distinct section** of a website while still being part of the main domain. It appears **before the main domain name** in a URL.

Example:

Let's say the main website is:

www.example.com

Now suppose the company wants to create a blog. Instead of making a new website, they can use a subdomain:

blog.example.com

Other possible subdomains:

- shop.example.com → for an online store
- support.example.com → for customer support
- login.example.com → for user login page

Why Use Subdomains?

Organize Content: Separate sections like blog, store, support, etc.

Different Technologies: Run a subdomain with a different CMS or app.

Multilingual Sites: Use subdomains for languages (e.g., fr.example.com for French).

Testing/Development: Use subdomains like dev.example.com or test.example.com.

10. Explain different types of ARP.

ARP is used to map an IP address to a MAC address within a local network. There are several **types of ARP**, each serving a specific purpose in network communication.

Proxy ARP:

- **Purpose**: Allows a router to respond to ARP requests on behalf of another machine.
- **Use Case**: Used when two devices are on different networks but appear to be on the same network.
- **Example**: A router replies with its own MAC address for a device on another network, making it reachable.

Gratuitous ARP:

- **Purpose**: Sent by a device to **announce or update its IP-to-MAC mapping**.
- **Use Case**: Used to detect IP conflicts or update other devices' ARP tables.
- **Example**: When a computer comes online, it sends a gratuitous ARP to inform others of its IP and MAC address.

Inverse ARP (InARP):

- **Purpose**: Opposite of ARP – it maps a **MAC address to an IP address**.
- **Use Case**: Used in Frame Relay and ATM networks.
- **Example**: A router knows the Layer 2 address but asks for the Layer 3 address.

Reverse ARP (RARP):

- **Purpose**: Resolves a **MAC address to an IP address**.
- **Use Case**: Used by diskless systems to obtain an IP address from a server.
- **Note**: Largely obsolete, replaced by **BOOTP** and **DHCP**.

Dynamic ARP:

- **Purpose**: Automatically updates ARP entries in the cache when a request/reply is sent or received.

- **Use Case**: Most common ARP behavior in day-to-day networking.

Static ARP:

- **Purpose**: Manually maps an IP address to a MAC address.
- **Use Case**: Useful for security, preventing spoofing or ensuring devices always use the same mapping.

11. Explain different type of DNS records.

Here's a clear explanation of the **different types of DNS (Domain Name System) records-**

A Record (Address Record):

- **Purpose**: Maps a **domain name** to an **IPv4 address**.
- **Use Case**: Directs users to your website server.

NS Record (Name Server Record):

- **Purpose**: Specifies the **authoritative DNS servers** for a domain.
- **Use Case**: Tells the internet where to find your domain's DNS zone.

TXT Record (Text Record):

- Purpose: Stores text data for various purposes.
- Common Use Cases:
  - Email verification
  - Site ownership verification
  - Custom metadata

CNAME Record (Canonical Name Record):

- **Purpose**: **Aliases** one domain name to another.
- **Use Case**: Useful for pointing subdomains to external services.

12. Write down about different FTP types.

Here is a clear explanation of the **different types of FTP (File Transfer Protocol)-**

Anonymous FTP:

- **Description**: Allows users to access files on an FTP server **without a username or password**.
- **Access**: Public; often used for downloading free resources like software or documentation.
- **Example Use**: Downloading files from a public university or open-source software repository.
- **Security**: Not secure — no encryption or authentication.

Password-Protected FTP:

- **Description**: Requires a **username and password** to log in.
- **Access**: Restricted to authorized users.
- **Example Use**: Website owners uploading site files to their hosting server.
- **Security**: Still insecure — data (including login info) is sent in **plain text**.

FTP Secure (FTPS):

- **Description**: FTP over **SSL/TLS** encryption (like HTTPS).
- **How it works**: The standard FTP protocol is enhanced with SSL/TLS to encrypt the connection.

  Types:

        - **Implicit FTPS** – encryption is required from the start
        - **Explicit FTPS** – starts with plain FTP, then upgrades to SSL/TLS
    - **Security**: Secure — data and credentials are encrypted.
    - **Port**: Uses port **990** (implicit) or **21** (explicit)

<u>FTP over Explicit SSL/TLS (FTPES):</u>
- **Description**: A version of FTPS where the client **explicitly requests SSL/TLS** encryption.
- **How it works**: The connection begins unencrypted and is **upgraded to encrypted** mode via command.
- **Flexibility**: Easier to configure behind firewalls because it uses the standard FTP port (21).
- **Security**: Secure — similar to FTPS but more flexible.
- **Port**: Uses port **21**

<u>Secure FTP (SFTP):</u>
- **Description**: Stands for **SSH File Transfer Protocol** — **completely different** from FTP.
- **Based on**: **SSH (Secure Shell)** protocol.
- **How it works**: Establishes a secure channel over port **22**, encrypting all commands and data.
- **Security**: Highly secure — preferred for confidential transfers.
- **Port**: Uses port **22**
- **Bonus**: Allows additional features like file permissions, symbolic links, etc.

13. Write down about the encoded data which is contained by a HTTP request.

When a client sends an **HTTP request** to a server, it can contain **encoded data** in various formats, depending on the request type (GET, POST, etc.). This data is used to send information like form submissions, authentication tokens, search queries, etc.

<u>**Common Parts of a HTTP Request That Contain Encoded Data-**</u>

<u>URL Query String (in GET requests):</u>
- Data is encoded in the URL after a '?' symbol.
- Key-value pairs are joined by '&' and spaces are replaced with '+' or '%20'.
- **Encoding type**: URL encoding (percent-encoding)

<u>Request Body (in POST, PUT, etc.):</u>
- The body can carry **form data**, **JSON**, **XML**, or **multipart** file uploads.
- The format is specified by the Content-Type header.

<u>Headers (like Cookies or Authorization):</u>
- Encoded data is also sent in headers.
- Authorization may use **Base64 encoding**.
- Cookie values are URL-encoded.

14. Describe different methods required for Network Security.

Network security is the practice of protecting a computer network and its data from unauthorized access, misuse, or theft. Below are the **key methods** used to secure networks:

<u>Firewalls:</u>
- **Purpose**: Act as a barrier between trusted internal networks and untrusted external networks (e.g., the internet).

- **Types**:
  - Software Firewalls (on devices)
  - Hardware Firewalls (on network perimeter)
- **Function**: Filters incoming and outgoing traffic based on defined rules.

Antivirus and Anti-Malware Software:

- **Purpose**: Detects and removes viruses, trojans, worms, ransomware, etc.
- **Function**: Scans files and systems in real-time or on demand to prevent malicious code execution.

Encryption:

- **Purpose**: Secures data by converting it into unreadable code.
- **Types**:
  - **Data-at-rest**: Stored files/data are encrypted.
  - **Data-in-transit**: Data traveling over the network (e.g., HTTPS).
- **Algorithms**: AES, RSA, SSL/TLS

Virtual Private Network (VPN):

- **Purpose**: Creates a secure tunnel over the internet for remote access.
- **Function**: Encrypts all data between the user's device and the VPN server.

Authentication Mechanisms:

- **Purpose**: Ensures only authorized users can access the network.
- **Methods**:
  - Username & Password
  - Two-Factor Authentication (2FA)
  - Biometric Authentication
  - Digital Certificates

Access Control:

- **Purpose**: Restrict access to sensitive parts of the network based on roles.
- **Types**:
  - Role-Based Access Control (RBAC)
  - Mandatory Access Control (MAC)
  - Discretionary Access Control (DAC)

15. Write down the characteristics of Selective Repeat Protocol (SRP) protocol.

Here are the key characteristics of the **Selective Repeat Protocol (SRP)**, a data link layer and transport layer protocol used for **reliable data transmission**:

Sliding Window Protocol:

- SRP is a **sliding window** protocol that allows the sender to send **multiple frames** (packets) before needing an acknowledgment.
- Both sender and receiver maintain a **window** of acceptable sequence numbers.

Selective Acknowledgment:

- Unlike Go-Back-N, SRP allows the receiver to **acknowledge individual frames**.
- If a frame is lost or arrives with errors, **only that frame is retransmitted**, not the entire window.

Out-of-Order Frame Handling:

- The receiver **accepts out-of-order frames** and buffers them until the missing frame is received.
- This improves **bandwidth utilization** and **efficiency** over unreliable networks.

Efficient Error Recovery:

- Since only erroneous or lost frames are resent, it reduces **redundant transmissions**, leading to better **network efficiency**.

Uses Sequence Numbers:

- SRP uses sequence numbers to keep track of sent and received frames.
- The sequence number space should be at least **twice the window size** to avoid ambiguity.

Requires More Memory and Complexity:

- Due to buffering and individual acknowledgment, SRP requires:
  - More **memory** for storing out-of-order frames
  - More **logic** for maintaining correct sequence

16. What is IP Masquerading? Write down its Pros & Cons.

IP Masquerading is a type of **network address translation (NAT)** used to allow multiple devices on a private network to **access the internet using a single public IP address**.

Pros of IP Masquerading:

Advantage	Description
Saves IP Addresses	Multiple devices can share one public IP — efficient use of IPv4 addresses.
Improved Security	Internal IPs are hidden from the internet, reducing the risk of direct attacks.
Cost-effective	Only one public IP is needed, reducing ISP charges.
Simplicity	Easy to set up on most routers and firewalls.

Cons of IP Masquerading:

Disadvantage	Description
Limits Inbound Traffic	External hosts cannot directly initiate a connection to internal devices.
Complicates Hosting Services	Hosting a server behind NAT requires additional configuration.
Breaks Some Protocols	Certain applications and protocols may not work properly without special NAT handling.
Debugging Difficulties	Makes network troubleshooting more complex due to address translation.

17. What is a Secure Socket Layer? How does SSL? Why is SSL Important?

SSL (Secure Socket Layer) is a **cryptographic protocol** designed to provide **secure communication** over the internet. It ensures **data encryption**, **authentication**, and **data integrity** between two communicating parties.

How Does SSL Work?

SSL works through a process called the **SSL Handshake**, which includes the following steps:

- Client Hello:

The browser sends a request to the server with supported **SSL versions**, **cipher suites**, and a **random number**.

- Server Hello:

  The server responds with:
  - Chosen **SSL version**
  - Selected **cipher suite**
  - Its **digital certificate (SSL certificate)** containing the **public key**
  - Another random number
- Certificate Verification:

The client **verifies the server's certificate** via a trusted Certificate Authority (CA).

- Session Key Creation:

The client and server **generate a session key** using the random numbers and public key via a secure algorithm like **Diffie-Hellman** or **RSA**.

- Secure Communication Starts:

- All communication is now **encrypted** using the session key.

Why is SSL Important?

Purpose	Description
Encryption	Prevents data from being read or altered by unauthorized parties.
Authentication	Confirms the identity of the website using certificates.
Data Integrity	Ensures data isn't modified or corrupted during transmission.
Protection Against Attacks	Protects against **Man-in-the-Middle (MITM)** attacks and eavesdropping.
Trust & SEO	HTTPS (secured by SSL) increases user trust and improves Google ranking.

18. What are cookies? Write down the basic uses. State about three different types of it with example.

Cookies are small pieces of data stored on a user's device by a web browser while browsing a website. They are used to **remember information** about the user, such as login sessions, preferences, or items in a shopping cart.

Basic Uses of Cookies:

Use Case	Description
Session Management	Maintain login status, track user sessions.
Personalization	Store user preferences, language, theme.
Tracking & Analytics	Monitor user behaviour, count visitors, etc.
E-commerce	Save cart items across pages or visits.

Types of Cookies (with Examples)-

Session Cookies:

- **Definition**: Stored temporarily and deleted when the browser is closed.
- **Purpose**: Maintain session state — such as keeping a user logged in as they navigate pages.
- **Example**:
  - When you log into your email and move between inbox and sent items, a session cookie keeps you authenticated.

Persistent Cookies:

- **Definition**: Stored on the user's device for a set period (days, months, or years).
- **Purpose**: Remember login details, preferences, or tracking info across sessions.
- **Example**:
  - “Remember me” checkbox on a login form stores login credentials so you don’t have to log in each time.

Third-Party Cookies:

- **Definition**: Set by a domain **different from the one the user is visiting**.
- **Purpose**: Track user activity across different websites, usually for advertising.
- **Example**:
  - Ads from Google or Facebook appearing on other websites set cookies to track your browsing habits.

19.Using HTML, CSS create a rotate effect for the image on hover.

Here's a simple example of how to create a **rotate effect on an image when hovered**, using HTML and CSS:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Rotate Image on Hover</title>
  <style>
    body {
      text-align: center;
      margin-top: 50px;
    }
```

```
  .rotate-image {
   width: 300px;
   transition: transform 0.5s ease;
  }

  .rotate-image:hover {
   transform: rotate(360deg);
  }
 </style>
</head>
<body>
 <h2>Hover Over the Image</h2>
 <img src="https://via.placeholder.com/300" alt="Sample Image" class="rotate-image">
</body>
</html>
```

20.What is UDP?

UDP (User Datagram Protocol) is a **connectionless** and **lightweight** transport layer protocol used for sending data over the internet. Unlike TCP, UDP **does not guarantee delivery, order, or error checking**, making it faster but less reliable.

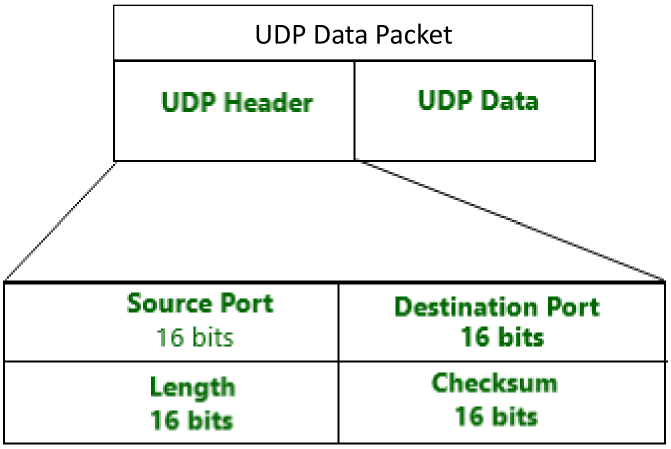
21.Describe UDP Packets with a suitable picture?

Here's a description of **UDP Packets** along with a visual representation of the **UDP Header** structure.

UDP Packet Structure:

A **UDP (User Datagram Protocol)** packet consists of a simple header and data section. The header is **8 bytes long** and includes four fields:

Field	Size	Description
Source Port	16 bits	Port of the sending application
Destination Port	16 bits	Port of the receiving application
Length	16 bits	Length of the UDP header + data
Checksum	16 bits	Error checking for header and data
Data	Variable	The actual payload/message being sent



22.What is the difference between SMTP, POP3, and IMAP.

Here's a clear comparison between **SMTP**, **POP3**, and **IMAP:**

Feature	SMTP	POP3	IMAP
Use	Sending mail	Receiving mail	Receiving mail
Direction	Outgoing only	Incoming only	Incoming only
Stores Emails On Server	Sends only	Deletes after download	Yes
Multi-device Access	No	No	Yes
Port (Default)	25	110	143
Secure Port	587 (TLS)	995 (SSL)	993 (SSL)

23.Write a javascript program to make the abbreviation of a given name as input.(Ex: Ramesh Chandra Das - R.C.Das)

Here's a simple **JavaScript program:**

```
<!DOCTYPE html>
<html>
<head>
  <title>Abbreviation Generator</title>
</head>
<body>
  <h2>Abbreviation Generator</h2>
  <input type="text" id="fullName" placeholder="Enter full name">
  <button onclick="makeAbbreviation()">Generate Abbreviation</button>
  <p id="result"></p>
```

```
 <script>
  function makeAbbreviation() {
    let name = document.getElementById("fullName").value.trim();
    let words = name.split(" ");
    if (words.length < 2) {
      document.getElementById("result").innerText = "Please enter at least a first and last
name.";
      return;
    }
    let abbr = "";
    for (let i = 0; i < words.length - 1; i++) {
      abbr += words[i].charAt(0).toUpperCase() + ".";
    }
    abbr += words[words.length - 1].charAt(0).toUpperCase() + words[words.length -
1].slice(1);
    document.getElementById("result").innerText = "Abbreviation: " + abbr;
  }
 </script>
</body>
</html>
```

24. Describe Class less Addressing with a suitable example. Demonstrate the benefits of it over Class-full address.

Classless Addressing is a method of IP address allocation and routing introduced to improve the limitations of **Classful Addressing**. It **does not divide IP addresses into fixed classes (A, B, C)** but instead allows for flexible division based on need using a **prefix notation.**

Example: 192.168.10.0/24

192.168.10.0 → Network address

/24 → The first 24 bits are the **network part**, remaining 8 bits are **host part**

<u>Benefits of Classless over Classful Addressing:</u>

Feature	Classful Addressing	Classless Addressing
IP Allocation	Fixed sizes (A: 16M, B: 65K, C: 256)	Flexible (based on need)
Network Utilization	Often wastes addresses	More efficient allocation
Subnetting	Limited and rigid	Supports variable-length subnetting
Routing	More entries in routing table	Route aggregation (supernetting) reduces size
Scalability	Poor for growing networks	Excellent scalability

25. Describe elaborately the life cycle of a packet through different layers of protocol it passes through when it traverse from Source to destination.

The **life cycle of a packet** as it travels from a **source to a destination** across a network follows the principles of the **OSI (Open Systems Interconnection) model**, or more commonly today, the **TCP/IP model**.

Application Layer:

- **Action**: User interacts with an application.
- **Data**: The application generates data.
- **Protocols**: HTTP, FTP, SMTP, DNS, etc.

Data is passed to the **Transport Layer**.

Transport Layer:

- **Action**: Breaks the data into segments (TCP) or datagrams (UDP).
- Adds **port numbers** to identify source/destination applications.
- **Protocols**: TCP (reliable), UDP (unreliable).

Segments are passed to the **Network Layer**.

Network Layer:

- **Action**: Adds **IP addresses** of source and destination.
- Determines the best **routing path** for the packet.
- **Protocols**: IP, ARP.

Encapsulated into **packets**, passed to the **Data Link Layer**.

Data Link Layer:

- **Action**: Converts packets into **frames**.
- Adds **MAC addresses** of the source and destination within the same network.
- Ensures **error detection** using CRC.
- **Protocols**: Ethernet, PPP, Frame Relay.

Frames are passed to the **Physical Layer**.

Physical Layer:

- **Action**: Converts frames into **electrical, optical, or radio signals**.
- Transmits bits (0s and 1s) over physical media.
- **Protocols**: IEEE 802.3 (Ethernet), IEEE 802.11 (Wi-Fi).

Packet leaves the source machine and travels over the network.

At the destination side the message again travels from the Physical Layer to the Application Layer.

26. Write Short notes on-

i. Go Back N ARQ

ii. TCP connection Life Cycle

iii. Inter Domain Routing

iv. Spanning Tree Broad

Here are **short and clear notes** on each of the topics-

i. Go-Back-N ARQ (Automatic Repeat reQuest):

- A **reliable data transmission protocol** used in networks.
- The sender can send **N frames** without waiting for an acknowledgment (using a sliding window).

- If any **frame is lost or corrupted**, the **receiver discards it** and all subsequent frames, even if they were correct.
- The **sender goes back** and retransmits from the **lost frame onwards**.
- **Drawback**: Wastes bandwidth because even correctly received frames are resent.

Example:
If frame 3 is lost in a 5-frame window, frames 3, 4, and 5 will all be resent.

ii. <u>TCP Connection Life Cycle:</u>

- TCP is **connection-oriented**. Before sending data, it establishes a connection using a **3-way handshake**:
    1. **SYN**: Client sends a synchronization request to the server.
    2. **SYN-ACK**: Server acknowledges and sends its own SYN.
    3. **ACK**: Client acknowledges server's response.

→ **Connection Established**

- **Termination** is done via a **4-step process** using **FIN** and **ACK** packets to gracefully close the connection from both sides.

iii. <u>Inter-Domain Routing:</u>

- Refers to **routing between different autonomous systems (AS)** on the internet.
- Each AS is a network or group of networks under a single administrative domain.
- Uses **Border Gateway Protocol (BGP)** to exchange routing information between domains.
- BGP decides the best route using policies, path attributes, and not just shortest path.

iv. <u>Spanning Tree Protocol (STP):</u>

- A **network protocol** used in Ethernet networks to **prevent loops**.
- Without STP, redundant paths can cause broadcast storms and multiple frame copies.
- STP elects a **Root Bridge** and blocks **redundant paths**, forming a loop-free **spanning tree**.
- Ensures that there is **only one active path** between two devices.

Used in: Switches and bridges in Layer 2 (Data Link Layer) networks.

27. Write a JavaScript program to divide a given array of positive integers into two parts. First element belongs to the first part, second element belongs to the second part, and third element belongs to the first part and so on. Now compute the sum of two parts and store it in an array of size two.

Here's a **JavaScript program:**

```
<!DOCTYPE html>
<html>
<head>
  <title>Alternate Sum Split</title>
</head>
<body>

  <h2>Alternate Array Split & Sum</h2>
  <script>
    function splitAndSum(arr) {
      let sum1 = 0;
```

```
    let sum2 = 0;
    for (let i = 0; i < arr.length; i++) {
      if (i % 2 === 0) {
        sum1 += arr[i];
      } else {
        sum2 += arr[i];
      }
    }

    return [sum1, sum2];
  }

  const inputArray = [10, 20, 30, 40, 50, 60];
  const result = splitAndSum(inputArray);
  document.write("<strong>Input Array:</strong> [" + inputArray.join(", ") + "]<br>");
  document.write("<strong>Result [Part1 Sum, Part2 Sum]:</strong> [" + result.join(", ") +
"]");
 </script>
</body>
</html>
```

28.A) With the help of the picture of Internet Protocol Version 4 (IPv4) Datagram Format describe at least 5 fields along with its functionalities.
B) Write down the difference between IPV4 and IPV6 datagram formats.

A) Here's the standard **IPv4 header format:**

Version (4 bits):

- Indicates the IP version.
- For IPv4, the value is **4**.
- Helps routers understand how to interpret the datagram.

Total Length (16 bits):

- Specifies the **entire size of the packet** (header + data) in bytes.
- Max size = 65,535 bytes.
- Essential for packet reassembly and transmission.

Identification (16 bits):

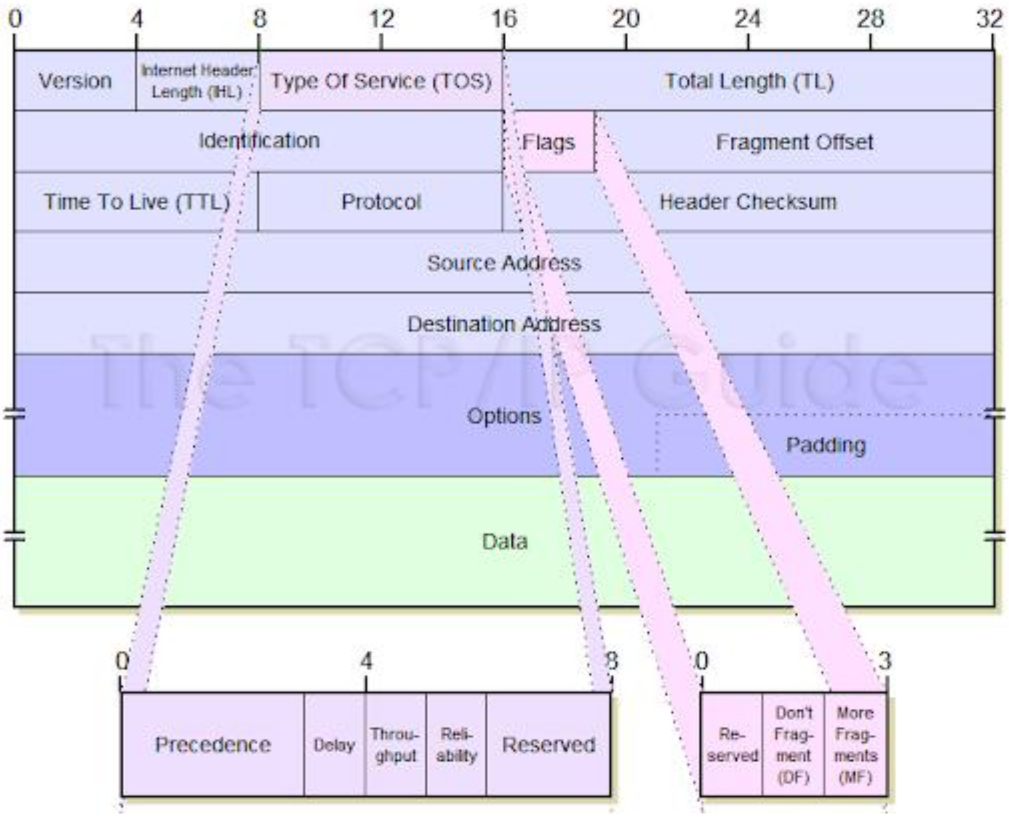
- A unique number for each datagram sent.
- Used for **fragmentation and reassembly** when a large packet is broken into smaller ones.

Time to Live (TTL) (8 bits):

- Limits the lifetime of the packet in the network.
- Each router it passes through **decreases TTL by 1**.
- If TTL = 0, packet is discarded. Prevents **infinite loops**.

Destination IP Address (32 bits):

- The final **receiving address** of the datagram.
- Crucial for packet delivery to the correct recipient.



B) Differences Between IPv4 and IPv6 Datagram Formats:

Feature	IPv4	IPv6
Header Size	20–60 bytes (variable)	40 bytes (fixed)
IP Address Length	32-bit	128-bit
Fragmentation	Done by sender and routers	Done **only by sender**
Header Checksum	Present	**Removed**
Options	Optional, variable-length	Handled using **Extension Headers**
Fields	Many fields, more complex parsing	Simplified header with fewer fields
Broadcast	Supported	**No broadcast**
Security	Optional (IPSec)	**Mandatory** support for IPSec

29. Define Routing. Elaborate different types of Routing with example along with its advantages and limitations.

Routing is the process of selecting a path over which data is sent from the source to the destination in a network. Routers determine the best route for forwarding packets based on routing algorithms and routing tables.

Types of Routing-

Routing can be broadly classified into **three main types**:

- Static Routing:

In static routing, routes are **manually configured** by a network administrator and do not change unless manually updated.

Example:

A small office network where the admin sets the default route to the ISP manually.

Advantages:

- Simple to configure in small networks.
- Less overhead; no bandwidth consumed for routing updates.
- More secure as routing paths don't change dynamically.

Limitations:

- Not scalable for large networks.
- No automatic adjustment if a link fails.
- Requires manual intervention to update routes.

- Dynamic Routing:

In dynamic routing, routers **automatically learn and update** routes using routing protocols.

Example:

Large enterprise networks using **OSPF (Open Shortest Path First)** to find the best path dynamically.

Advantages:

- Scales well for large networks.
- Automatically adapts to topology changes (e.g., failed links).
- Reduces manual configuration errors.

Limitations:

- Consumes bandwidth for route updates.
- More complex to configure.
- Routing loops may occur without proper configuration.

- Default Routing:

All traffic for **unknown destinations** is sent to a **default router** (gateway). Often used in networks with only one exit path.

Example:

A home router sending all non-local traffic to the ISP.

Advantages:

- Easy to configure.
- Useful in stub networks with only one way out.

Limitations:

- Not suitable for networks with multiple external paths.
- Can create bottlenecks if overused.

30. Write javascript code for set, get and check a cookie for a website. Write a JavaScript code to Count Frequencies of Array Element.

<u>JavaScript Code to Set, Get, and Check a Cookie:</u>

```
<!DOCTYPE html>
<html>
<head><title>Cookie Example</title></head>
<body>

<script>
function setCookie(name, value, days) {
  let expires = "";
  if (days) {
    const date = new Date();
    date.setTime(date.getTime() + (days*24*60*60*1000));
    expires = "; expires=" + date.toUTCString();
  }
  document.cookie = name + "=" + value + expires + "; path=/";
}

function getCookie(name) {
  const cname = name + "=";
  const decodedCookie = decodeURIComponent(document.cookie);
  const ca = decodedCookie.split(';');
  for(let i = 0; i < ca.length; i++) {
    let c = ca[i].trim();
    if (c.indexOf(cname) === 0) return c.substring(cname.length, c.length);
  }
  return "";
}

function checkCookie() {
  const user = getCookie("username");
  if (user !== "") {
    alert("Welcome back, " + user + "!");
  } else {
    const name = prompt("Please enter your name:");
    if (name !== "" && name !== null) {
      setCookie("username", name, 7);
    }
  }
}

checkCookie();
</script>
</body>
</html>
```

JavaScript Code to Count Frequencies of Array Elements:

```
function countFrequencies(arr) {
  const freq = {};
  for (let i = 0; i < arr.length; i++) {
    const item = arr[i];
    freq[item] = (freq[item] || 0) + 1;
  }
  return freq;
}

const inputArray = [1, 2, 2, 3, 1, 4, 2, 3];
const frequencyMap = countFrequencies(inputArray);
console.log("Frequency of elements:", frequencyMap);
```

31. Write a JavaScript Program to Cyclically Rotate an Array by one.

Here's a simple and efficient **JavaScript program** to **cyclically rotate an array by one.**

```
function rotateArrayByOne(arr) {
  if (arr.length === 0) return arr;
  const last = arr.pop();
  arr.unshift(last);
  return arr;
}

let arr = [1, 2, 3, 4, 5];
console.log("Original Array:", arr);
rotateArrayByOne(arr);
console.log("Rotated Array:", arr);
```

32. Write code to show How to Zoom an Image on Mouse Hover using CSS.

Here's a simple example of **how to zoom an image on mouse hover using CSS**:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Image Zoom on Hover</title>
  <style>
    .image-container {
      width: 300px;
      height: 200px;
      overflow: hidden;
      border: 2px solid #333;
    }

    .image-container img {
      width: 100%;
      height: 100%;
```

```
    transition: transform 0.3s ease;
  }

  .image-container:hover img {
    transform: scale(1.2);
  }
 </style>
</head>
<body>
 <h2>Hover to Zoom the Image</h2>
 <div class="image-container">
  <img src="https://via.placeholder.com/300x200" alt="Zoom Image">
 </div>
</body>
</html>
```

33. Difference between Search Engine and Web Browser.

Here is a clear comparison between a **Search Engine** and a **Web Browser**:

Feature	Search Engine	Web Browser
Definition	A tool used to search for information on the Internet.	A software application used to access and view websites.
Purpose	To help users find websites or information.	To retrieve, display, and navigate web pages.
Examples	Google, Bing, Yahoo, DuckDuckGo	Chrome, Firefox, Edge, Safari, Opera
How it works	Uses bots and indexing to display relevant web results.	Uses URLs or search engines to open and display web pages.
Requires Internet?	Yes	Yes
User Interface	Search bar, results page	Address bar, tabs, bookmarks, back/forward navigation
Installed on?	Not installed, accessed via a browser	Installed on devices as an application

34. Write down the difference Between Web server and Application server.

Here is a clear comparison between a **Web Server** and an **Application Server**:

Feature	Web Server	Application Server
Purpose	Serves static content.	Serves dynamic content by executing business logic.
Primary Role	Handles HTTP requests and responses.	Provides application-level services.
Content Type	Static content	Dynamic content
Processes	HTTP requests only	HTTP + additional protocols
Examples	Apache HTTP Server, Nginx	Apache Tomcat, JBoss, WebLogic, GlassFish
Technology Support	HTML, CSS, JS	Java, .NET, PHP, Python, etc.
Performance	Fast for static content	Slower due to logic processing
Integration	Can forward requests to app servers	Often integrates with databases, web servers, and services

35. How to reset/remove CSS styles for element.

To **reset or remove CSS styles** for an element, you can use one or more of the following methods:

- Use all: unset –

```
.element {
  all: unset;
}
```

Effect: Resets all inherited and non-inherited styles to their initial or inherited state.
Use case: Useful when you want to remove **all default browser and custom styles**.

- Use initial or inherit for individual properties -

```
.element {
  color: initial;
  font-size: inherit;
}
```

initial → Resets to browser default.
inherit → Takes value from the parent element.

- Use a CSS Reset -

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}
```

Removes margin/padding inconsistencies across browsers.

Often included at the start of a CSS file.

- Use revert -

```
.element {
  all: revert;
}
```

Effect: Reverts the property to the value it would have had from user-agent stylesheets.
Better than initial in some cases, especially in forms.

36. How to create Table with width, 100% with Vertical Scroll inside Table body in HTML.

HTML + CSS Code: Scrollable <tbody> with Full-Width Table -

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Scrollable Table Body</title>
  <style>
    table {
      width: 100%;
      border-collapse: collapse;
    }

    thead th {
      background-color: #333;
      color: white;
      position: sticky;
      top: 0;
      z-index: 2;
    }


th, td {
      border: 1px solid #ccc;
      padding: 8px;
      text-align: left;
    }

    .table-container {
      max-height: 200px;
      overflow-y: auto;
      display: block;
    }

    tbody {
      display: block;
      width: 100%;
```

```
    }

    thead, tbody tr {
      display: table;
      width: 100%;
      table-layout: fixed;
    }
  </style>
</head>
<body>
  <h3>Full-width Table with Scrollable Body</h3>
  <div class="table-container">
    <table>
      <thead>
        <tr>
          <th>Sl No</th>
          <th>Name</th>
          <th>Email</th>
        </tr>
      </thead>
      <tbody>
        <tr><td>1</td><td>John Doe</td><td>john@example.com</td></tr>
        <tr><td>2</td><td>Jane Smith</td><td>jane@example.com</td></tr>
        <tr><td>3</td><td>Mike Ross</td><td>mike@example.com</td></tr>
        <tr><td>4</td><td>Rachel Zane</td><td>rachel@example.com</td></tr>
        <tr><td>5</td><td>Louis Litt</td><td>louis@example.com</td></tr>
        <tr><td>6</td><td>Harvey Specter</td><td>harvey@example.com</td></tr>
        <tr><td>7</td><td>Donna Paulsen</td><td>donna@example.com</td></tr>
      </tbody>
    </table>
  </div>
</body>
</html>
```